Planrs Academy

FRICTION — BRAKE PADS INVESTIGATION

HAS ANYONE PRESSED BICYCLE BRAKES?

Has anyone ever been on a bicycle and pressed the brakes? What happened? WHY did it stop?

Think about what you felt when the bicycle slowed down.

Share your experience with your classmates!

SPOT THE FRICTION!

Look at these three pictures carefully. Can you find what is the SAME in all of them?

A child pressing bicycle brakes to stop

Two hands rubbing together quickly

A cricket ball rolling and slowing down

WHAT IS FRICTION?

Friction is the force that happens when two things rub against each other. It slows moving things down or stops them. Imagine sliding down a playground slide — friction is what helps slow you down!

What happens when things rub together!

ROUGH VS. SMOOTH SURFACES

Some surfaces grip more, some surfaces slip more. Let's discover how friction works differently on rough and smooth surfaces!

ROUGH Surfaces: Grip more → More friction → Stops faster!

SMOOTH Surfaces: Slip more → Less friction → Keeps moving!

Examples: sandpaper, road surface, rubber shoe sole

Examples: glass, marble floor, plastic sheet

INDIAN EXAMPLES — ROUGH VS SMOOTH

Friction is all around us in India! Let's look at everyday things we use that are rough or smooth.

Rubber chappal on wet floor — grips well!

Polished marble temple floor — smooth and slippery!

Coir doormat and cycle tyre tread — rough surfaces!

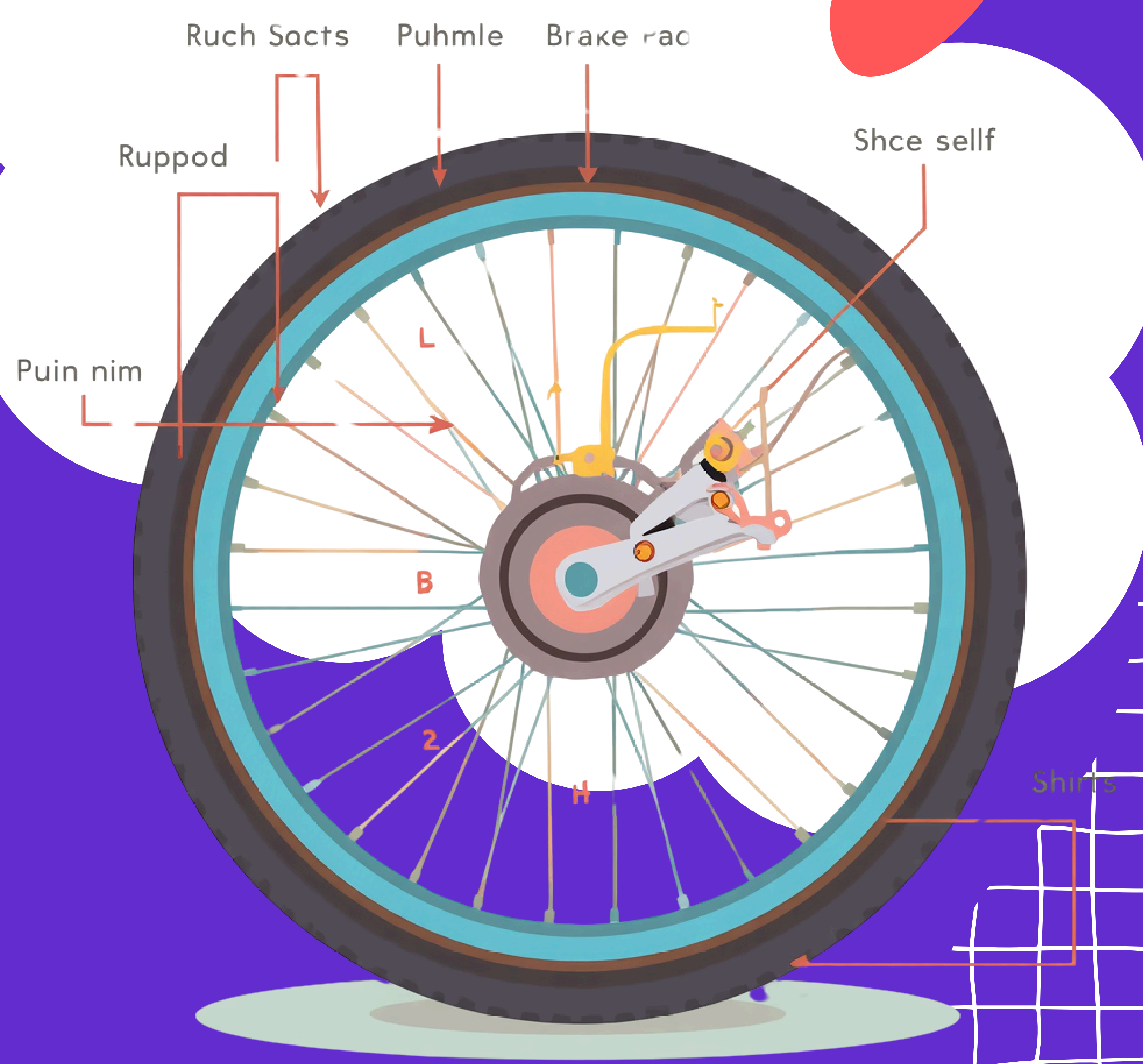
Glass mobile screen and steel tumbler — smooth surfaces!

WHAT ARE BRAKE PADS?

Look Inside a Bicycle Wheel

Brake pads are rough pads inside wheels. When you press the brakes, these rough pads press against the wheel to stop it!

Rough pads create friction to slow down the wheel!



HOW BRAKE PADS WORK

When you want to stop your bicycle, brake pads use friction to help you! Follow these 3 simple steps to see how it works.

Step 1: You squeeze the brake handle with your hand.

Step 2: The rough brake pad presses against the spinning wheel.

Step 3: LOTS of friction is created → wheel slows → bicycle stops → You are safe!

INVESTIGATION TIME!

Which Surface Stops the Block Fastest? Let's find out using friction! Gather your materials and get ready to experiment.

- Materials: wooden block or tiffin box
- 4 surfaces: newspaper, sandpaper, foil, cloth
- Ramp (textbook) • Be careful and gentle!

STEP 1 - PREDICT

Look at the 4 surfaces: newspaper, sandpaper, aluminium foil, and cloth. Which one do you think will stop the block the fastest? Think about how each surface feels - is it rough or smooth?

I think the _________ surface will stop the block fastest because it looks ________.

STEP 2 - TEST

Now it's time to test each surface! Follow these simple steps to see which surface creates the most friction.

Place each surface at the bottom of your ramp.

Slide the block down the ramp gently.

Mark where the block stops on each surface!

OBSERVE & RECORD

Now let's record what happened! Fill in the table below for each surface you tested. Was it rough or smooth? Did the block stop near the ramp or far away? Compare your results with your friends!

Draw what you observed in your notebook!

DISCUSSION QUESTIONS

Think and Share!

Let's talk about what we discovered in our friction experiment. Share your thoughts with your classmates!

Q1: Which surface stopped the block fastest? Why?

Q2: Which surface is most like a brake pad — sandpaper or foil? Why?

Q3: Where does friction keep YOU safe at home or on your street?

INDIAN EXAMPLES — FRICTION KEEPS US SAFE!

Friction is all around us in India! Let's see how it keeps us safe every day in our homes, schools, and streets.

Bata school shoe sole grips the staircase steps.

Auto-rickshaw brake pads stop safely at school gate.

Cricket bat rubber grip stops hands from slipping.

Rubber chappal sole grips wet bathroom floor.

3 BIG THINGS WE LEARNED TODAY

Key Takeaways

Let's remember the important things we discovered about friction today!

1. Friction happens when two surfaces rub together
2. Rough surfaces make MORE friction — they stop things faster
3. Brake pads are rough on purpose — they use friction to stop wheels!

EXIT TICKET — THUMBS UP OR DOWN?

Show thumbs up for MORE friction, thumbs down for LESS friction! Look at each object and decide.

Rough Sandpaper — lots of tiny bumps!

Smooth Plastic Sheet — shiny and slippery!

Rubber Chappal Sole — grippy texture!

TAKE-HOME CHALLENGE

Friction Detective Homework

Find 3 things at home that use friction to keep you safe. Examples: doormat, rubber handle on vessel, slippers, bicycle brakes, eraser.

Draw them and bring your drawings tomorrow!

WE ARE FRICTION DETECTIVES!

Key Words We Learned

Today we learned about FRICTION — the force that happens when things rub together. We discovered how ROUGH and SMOOTH surfaces work differently!

FRICTION • ROUGH • SMOOTH • BRAKE PAD • GRIP

THANK YOU